

PHYSICS 550
Homework #7
Due on Nov 27, 2023

1. G 4.21 (10 points)
2. G 4.29 (10 points)
3. G 4.36 (10 points)
4. G 4.38 (10 points)
5. (10 points) 3 identical, spinless bosons of mass m are noninteracting and they each reside in an energy eigenstate of a one-dimensional simple harmonic oscillator potential of frequency ω . It is known that they are distributed among the following 3 orbitals: 1 boson is in the ground state orbital of the oscillator ($n=0$) and 2 bosons are in the second excited state orbital ($n=2$) of the oscillator. Answer the following:
 - (a) What are the total energy and normalized state vector of this 3-boson system?
 - (b) Calculate the expectation value of the following operator, $Q = (x_1)^2 + (x_2)^2 + (x_3)^2$